

Use the formula for the sum of the first n integers and/or the formula for the sum of a geometric sequence to evaluate the sums in 20-29 or to write them in closed form.

5.2.26. $3 + 3^2 + 3^3 + \dots + 3^n$, where n is any integer with $n \geq 1$.

Theorem of sum of a Geometric Sequence:

$$\sum_{i=0}^n r^i = \frac{r^{n+1} - 1}{r - 1}$$

$$r = 3$$

$$i = 1$$

$$\sum_{i=1}^n 3^i = 3^{(1)} + 3^{(2)} + 3^{(3)} + \dots + 3^{(n)}$$

$$\left(\frac{3^{n+1} - 1}{3 - 1} \right) - 1 = \left(\frac{3^{n+1} - 1}{2} \right) - 1$$

Solution: $\sum_{i=1}^n 3^i = \left(\frac{3^{n+1} - 1}{3 - 1} \right) - 1$